
Cryptography (02-24-06203)

Course Presentation

by

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Course Outline

Prerequisite :

**02-24-06201 (Introduction to Cybersecurity),
02-24-06202 (Number Theory)**

Topics:

**Course Overview
Introduction to Cryptography
Data Encryption Standard (DES)
Advanced Encryption Standard (AES)
Public Key Cryptography
Network Layer Security: IPSec
Transport Layer Security: SSL, TLS
Application Layer Security: E-Mail Security
Steganography**

Grading:

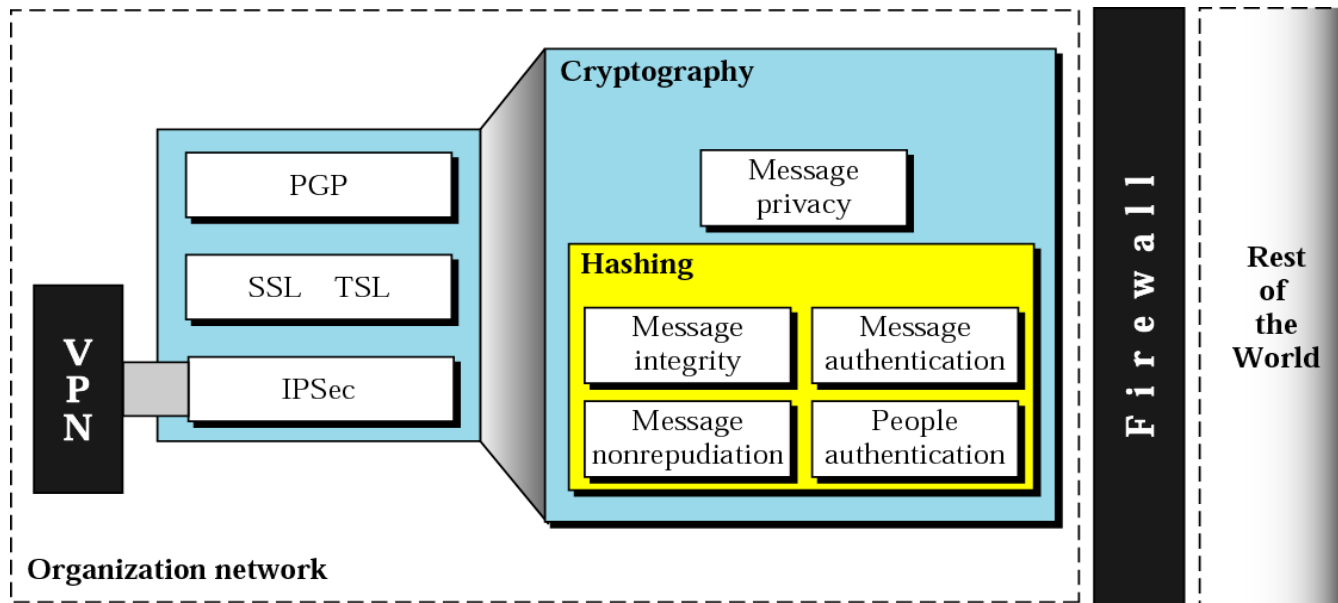
**Final Exam (50 %)
Lab Works (20 %)
Midterm Exam (30 %)**

Textbook:

**William Stallings, "Cryptography and Network Security: Principles and Practices,"
Pearson Education Limited, 8th Edition, ISBN 13: 978-1-292-43749-1, 2023.**

Course Overview

The Field of Security Roadmap



1. **Cryptography:** is the basic foundation and corner stone for all security topics in computer networks whose main purpose is to achieve message privacy,
2. **Hashing:** is a subfield in Cryptography whose purpose is to achieve: **Message Integrity**, **Message Authentication**, **Message Non Repudiation**, and **People Authentication**,
3. **PGP, SSL/TSL, or IPsec:** **Three levels of security** are available for computer networks, the first PGP (Pretty Good Privacy) is available at the application layer to secure eMail Messages, SSL/TSL is available at the Transport layer to secure TCP segments, and IPsec is available at the Network layer (through VPN) to secure IP datagrams,
4. **Firewalls:** used to protect the site, from Intruders or Malicious software, by having a router as a first line defense and is the only way to connect with the rest of the world.

The Cryptography Course

✓ **This course is organized into four broad categories:**

➤ **Cryptographic algorithms**

- ★ symmetric encryption,
- ★ asymmetric encryption,
- ★ hash functions

➤ **Internet Security,**

➤ **Network Security, and**

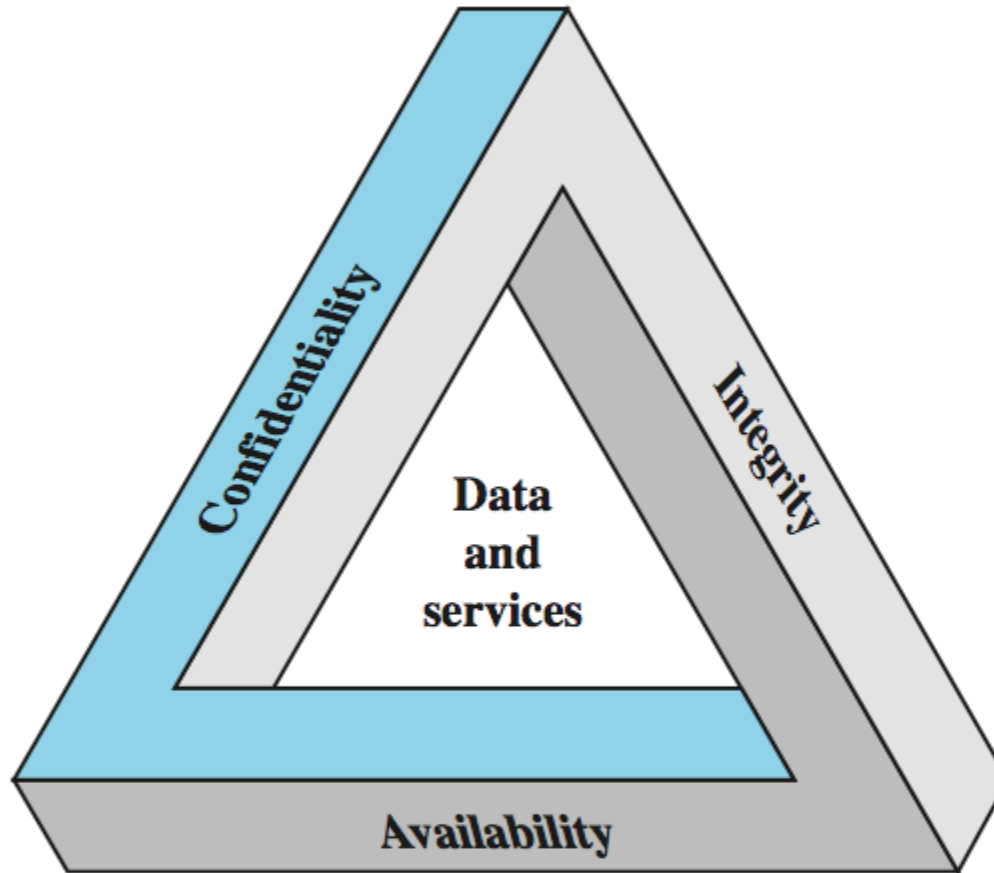
➤ **Computer Security**

Security Facts

Two fundamental security facts:

1. All complex software programs have flaw/bugs,
2. It is extremely difficult to build hardware/software not vulnerable to attack.

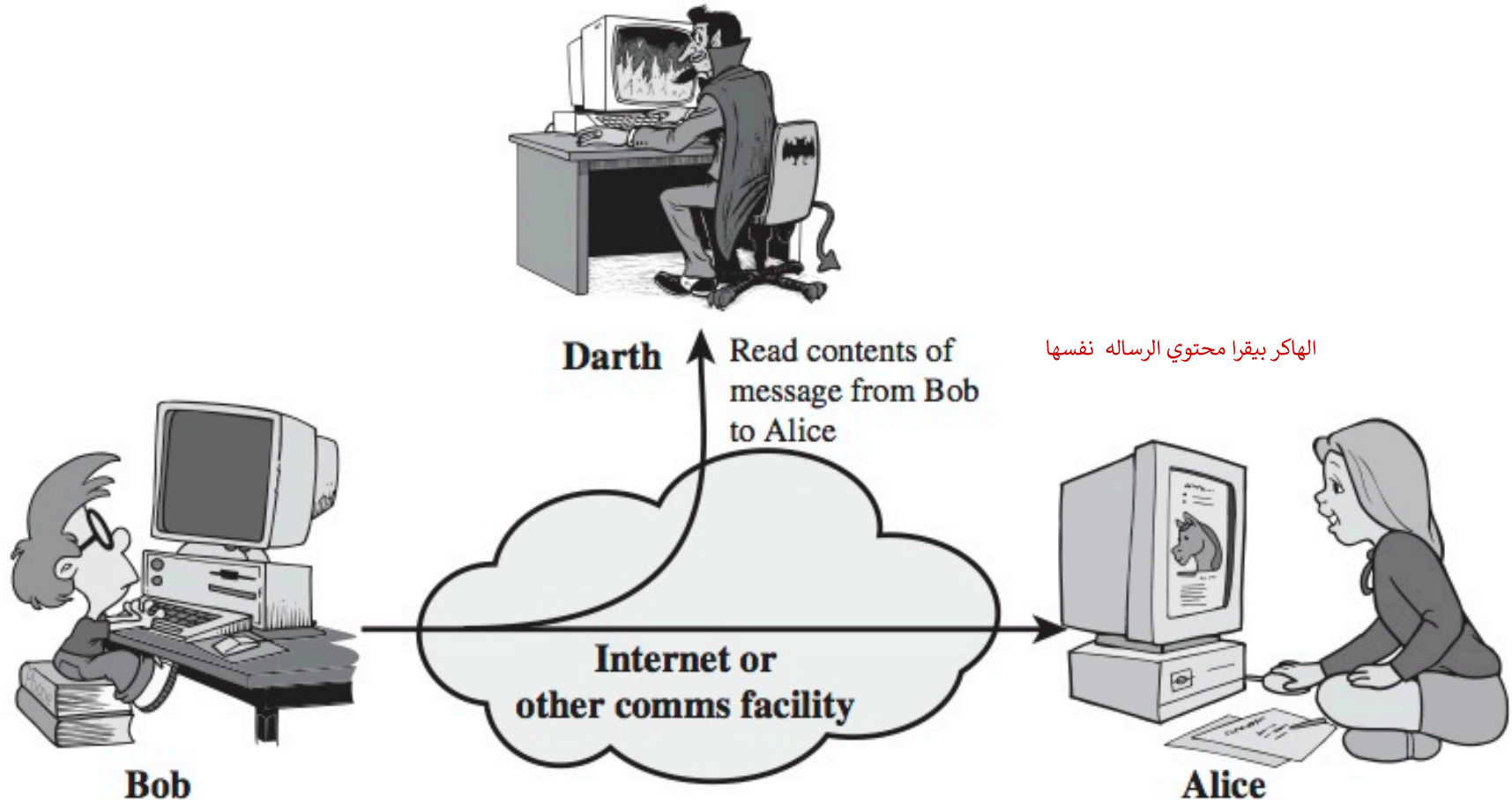
Key Security Concepts (The C.I.A. TRIAD)



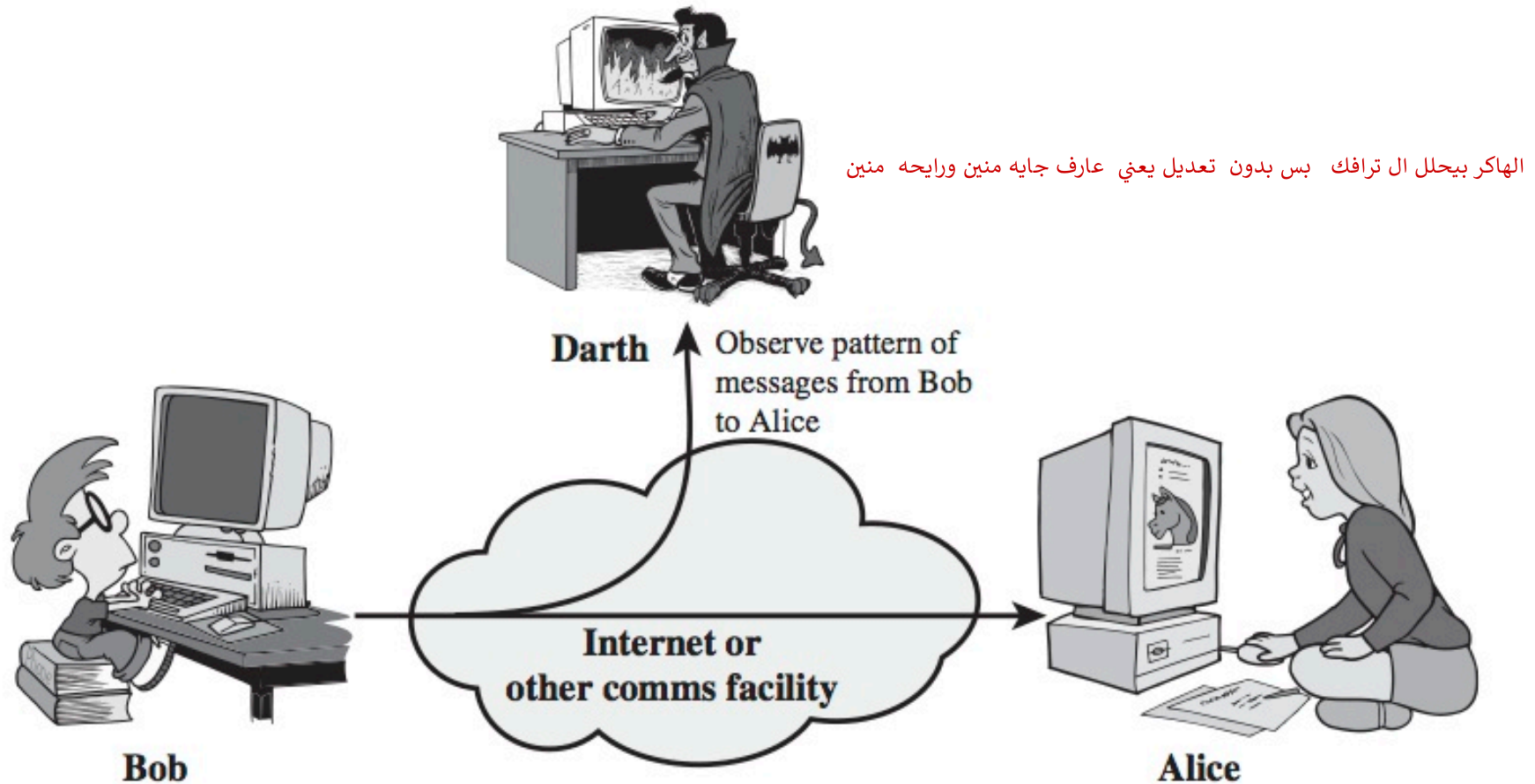
Passive Attacks: release of message contents

الهجمات السلبية

اصدار الرساله



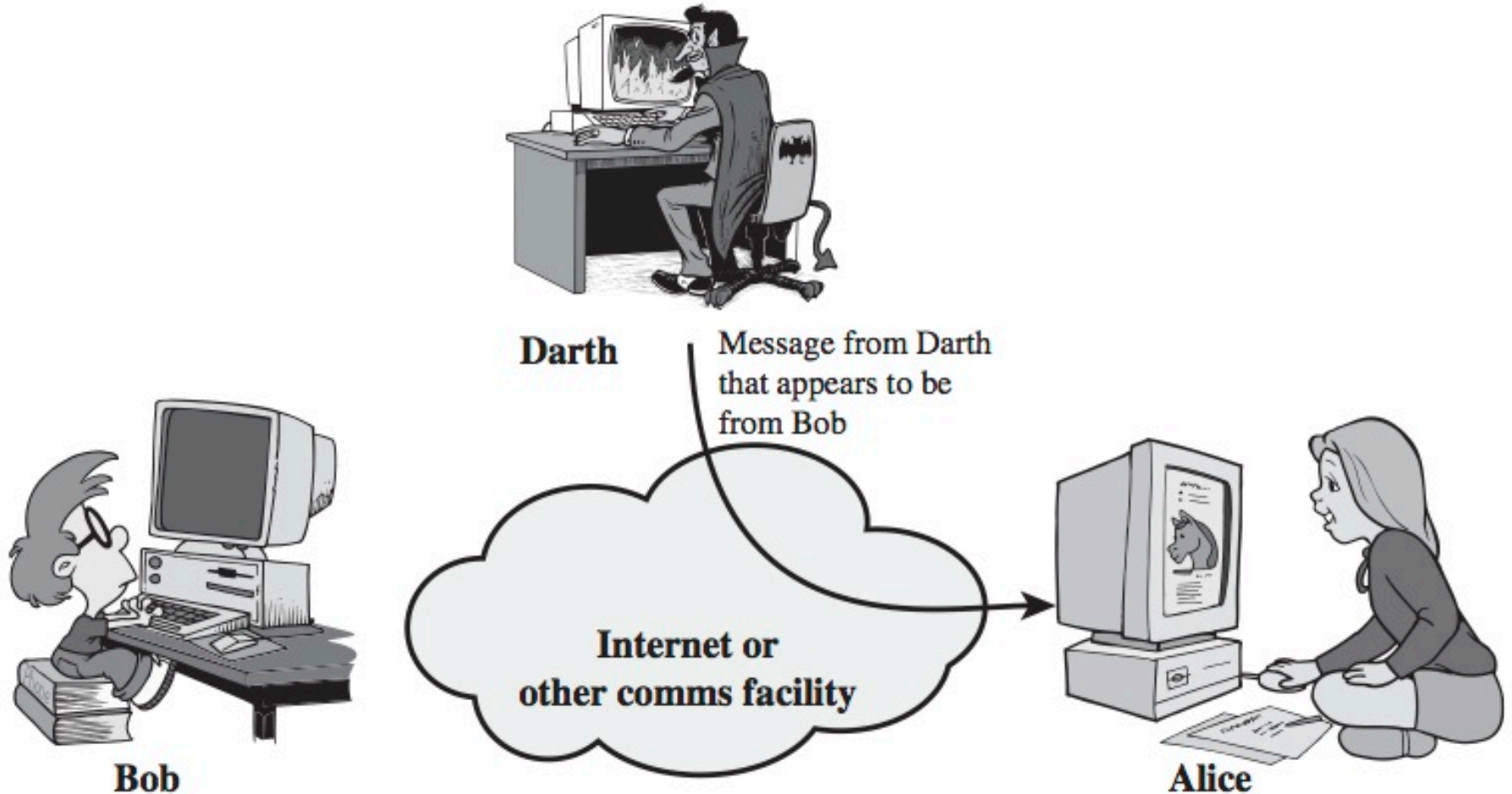
Passive Attacks: traffic analysis



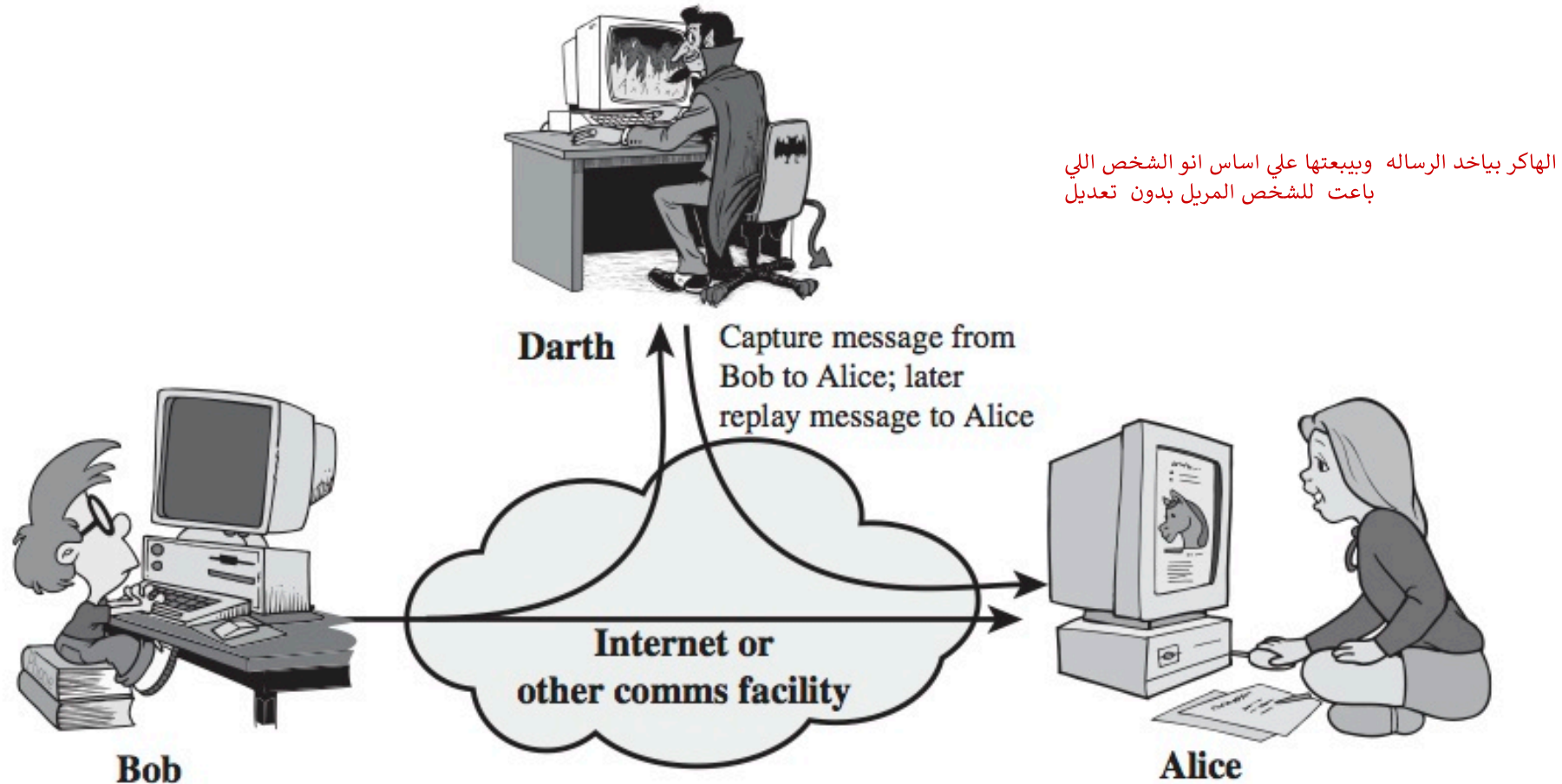
Active Attacks: masquerade

الهجمات النشطة

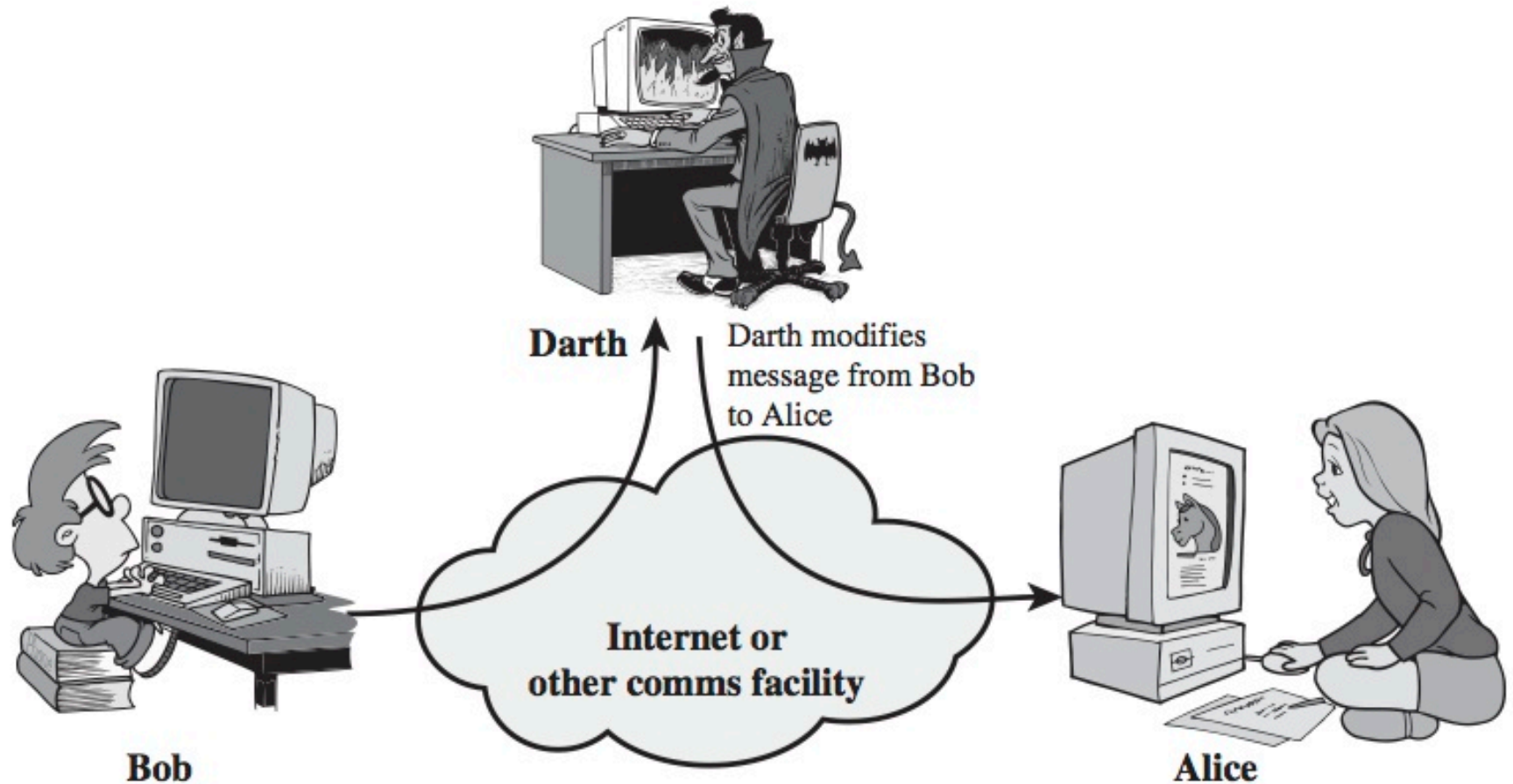
انتحال الهوية



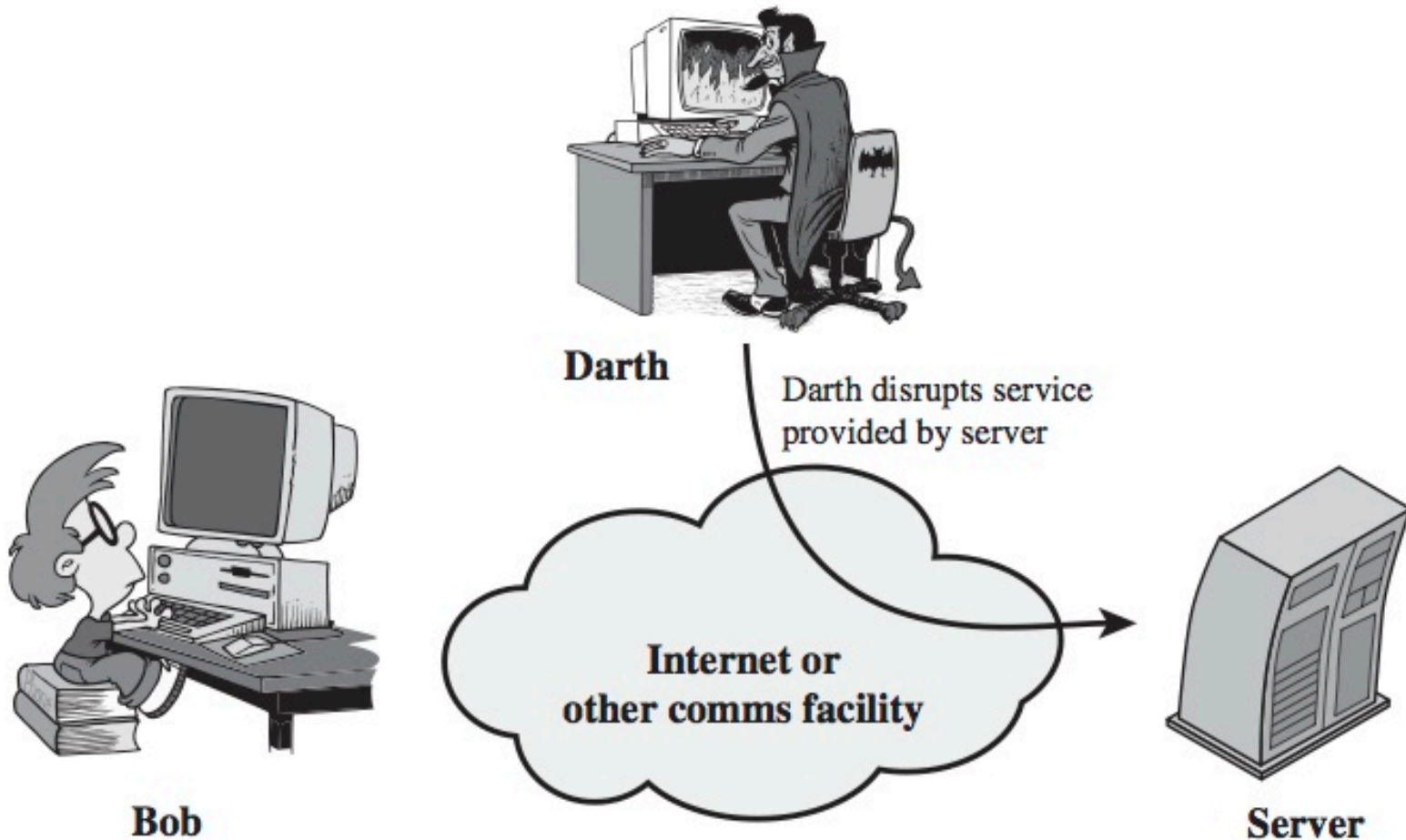
Active Attacks: replay



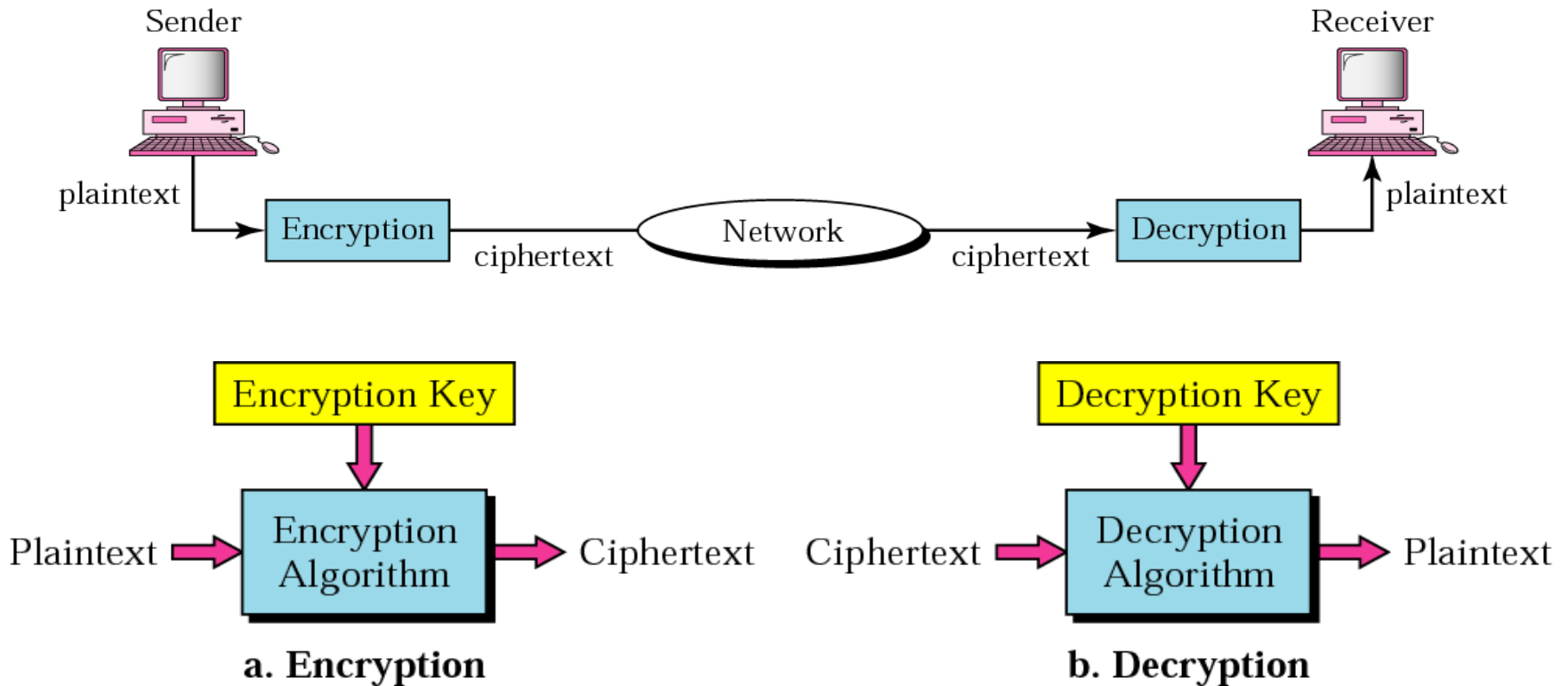
Active Attacks: modification of messages



Active Attacks: denial of service

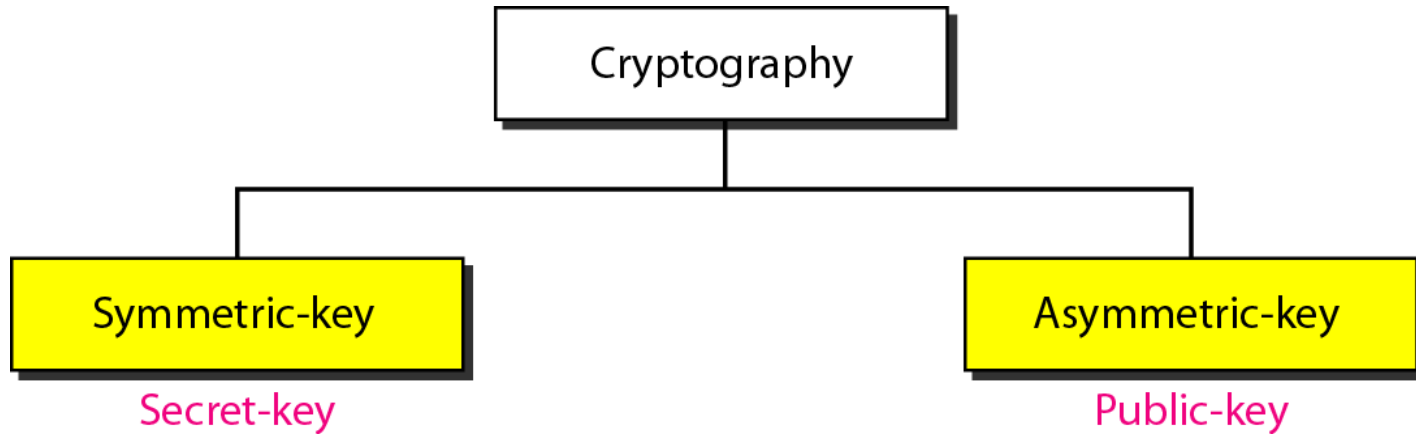


Cryptography Components

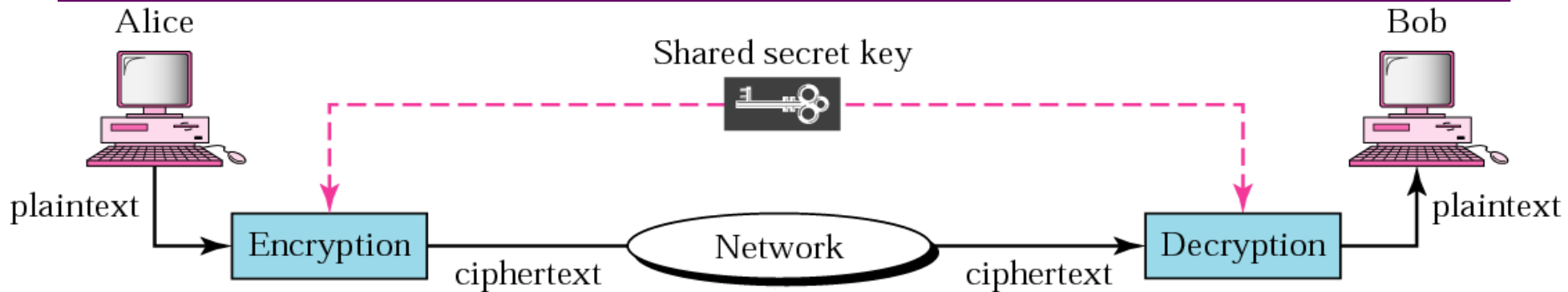


**In cryptography,
the encryption/decryption algorithms
are known; the keys are secret.**

Categories of Cryptography

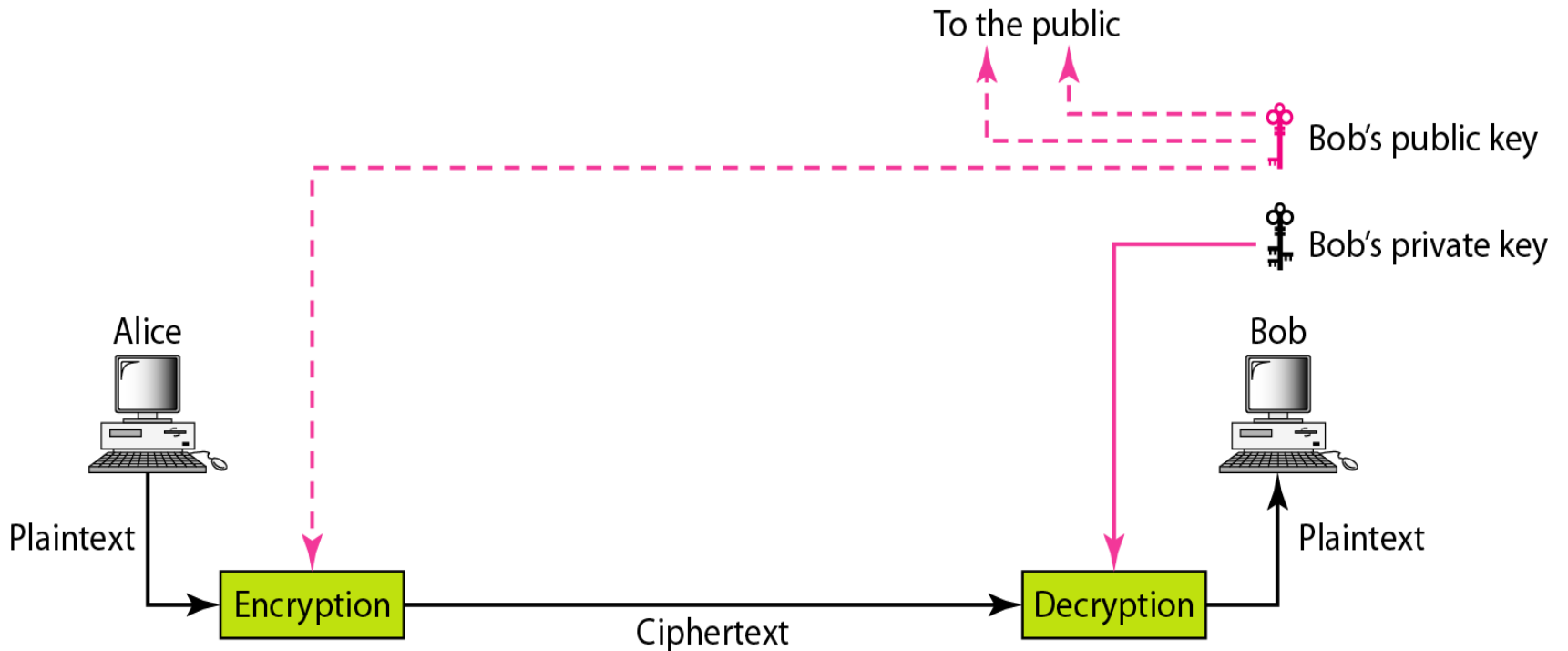


Symmetric-Key Cryptography



- 1. The same key used by the sender (encryption) will be used again by the receiver (decryption),**
- 2. Thus, the key is shared between the sender and the receiver,**
- 3. It is the recommended encryption for long messages.**

Asymmetric-Key Cryptography



Keys used in Cryptography



Secret key

Symmetric-key cryptography



Public key



Private key

Asymmetric-key cryptography

Some Basic Terminology

plaintext - original message

ciphertext - coded (encrypted) message

cipher - algorithm for transforming plaintext to ciphertext

key - info used in cipher known only to sender/receiver

encipher (encrypt) - converting plaintext to ciphertext

decipher (decrypt) - recovering ciphertext from plaintext

cryptography - study of encryption principles/methods

cryptanalysis (codebreaking) - study of principles/methods of deciphering ciphertext *without* knowing key

cryptology - field of both cryptography and cryptanalysis

Origin of Symmetric-Key Cryptography

Symmetric-key cryptography started thousands of years ago when people needed to exchange secrets (for example, in a war),

We still mainly use symmetric-key cryptography in our network security.

WiFi security algorithms evolved from the obsolete WEP, to WPA, to the robust WPA2, and finally WPA3 :

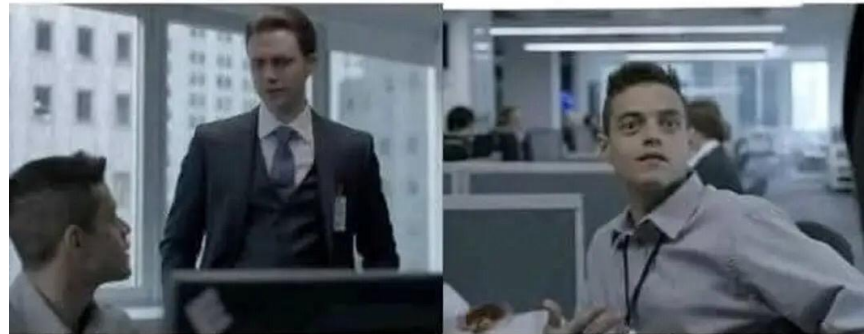
- 1. WPA3 is the most secure, providing data encryption and advanced, more secured password mechanism,**
- 2. WPA2 (using AES) remains a secure, widely used standard,**
- 3. WEP (using RC4) and WPA (using TKIP) are now deprecated due to severe vulnerabilities.**

From the Social Media (No Comments)



What is your address?

157.42.20.132



No, your local address?

127.0.0.1



I mean, your physical
address!

00:A0:C9:4F:73:2E

